

1. Load Word

LW is a 32-bit word from memory into a register. It requires ALU address computers, a memory read, and a register write.

Stage	Operation	Control Signal
IF	Instruction fetched, PC incremented $PC \leftarrow PC + 4; IR \leftarrow Mem[PC]$	PC Write = 1, IR Write = 1, MemRead = 1, ALUSrcB = 01, PC source = 00; MemWrite = 0, RegDst = 0; RegWrite = 0; MemtoReg = 0; ALUSrcA = 0; ALU Op = 00; Branch = 0
ID	Read rs, rt; compute branch target	PC Write = 0; PCWriteCond = 0; PC Source = 00; Iord = 0, MemRead = 0, MemWrite = 0; IR Write = 0, Reg Write = 0, RegDst = 0, MemtoReg = 0, ALUSrcA = 0, ALUSrcB = 11, ALU Op = 00, Branch = 0
EX	Effective address = A + sign - extended immediate	PC Write = 0, PCWriteCond = 0, PC source = 00, MemRead = 0, MemWrite = 0, IR Write = 0, IORD = 0, RegWrite = 0, RegDst = 0, MemtoReg = 0, ALUSrcA = 1, ALUSrcB = 10, ALU Op = 00, Branch = 0

MEM

Read Memory
 $MDR \leftarrow Mem[ALUout]$

$PCWrite = 0, PCWriteCond = 0,$
 $PCSource = 00, IorD = 1,$
 $MemRead = 1, MemWrite = 0,$
 $IRWrite = 0, RegWrite = 0, RegDst = 0,$
 $MemtoReg = 0, ALUSrcA = 0,$
 $ALUSrcB = 00, ALUOp = 00,$
 $Branch = 0$

WB

Write MDR to rt
 $Reg[rt] \leftarrow MDR$

$PCWrite = 0, PCWriteCond = 0, PCSource = 00,$
 $MemRead = 0, MemWrite = 0, IRWrite = 0,$
 $IorD = 0, RegWrite = 1, RegDst = 0,$
 $MemtoReg = 1, ALUSrcA = 0,$
 $ALUSrcB = 00, ALUOp = 00, Branch = 0$

Execution Flow

IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ MEM $\rightarrow$ WB

LN uses all 5 stages because it fetches the instructions, decodes it, computes the effective memory address, reads the word from memory and finally writes it back to destination register.

2 B Add (R-Type)

Add performs register to register addition and writes the result to `rd`

Stage	Operation	Control Signal
IF	Fetch Instruction	$PCWrite = 1, IRWrite = 1, MemRead = 1$ (Standard IF signals)
ID	Read registers <code>rs</code> and <code>rt</code> ;	$ALU SrcB = 11$, (Std. IF signal)
EX	Perform $(A+B)$	$ALU SrcA = 1, ALU SrcB = 00, ALU Op = 10$
MEM	No memory access	ALL signals 0
WB	$Reg[rd] \leftarrow ALU Out$	$RegWrite = 1, RegDst = 1, MemtoReg = 0$, Others 0

Execution Flow

IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ WB

Add uses the ALU in the EX stage and then writes the result to the destination register in WB. MEM is unused.

3 B Bne (Branch Not Equal)

Stage	Operation	Control Signal
IF	Fetch Instruction	PCWrite=1, IRWrite=1, MemRead=1, (standard IF signals)
ID	Read registers, compute branch target	ALUSrcA=0, ALUSrcB=11, ALUOp=00, all others 0
EX	compare rs - rt	ALUSrcA=1, ALUSrcB=00, ALUOp=01, PCWriteCond=1, Branch=1; all others 0
MEM	Branch decision applied If zero = 0 $\rightarrow$ PC $\leftarrow$ ALUOut	PCWriteCond=1, Branch=1, PCSource=01,
WB	No Write Back	RegWrite=0; all others 0

Execution Flow

IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ MEM

Bne evaluates the branch condition in EX and uses the MEM stage to update the PC if cond. is true

4B Jump

Jump replaces the PC with the jump target constructed from the instruction

Stage	Operation	Control Signal
IF	Fetch Instruction	standard IF signals
ID	Jump target compute PC update	PC Write = 1, PC Source = 10 all others 0
EX	No operation	All signals 0
MEM	No operation	All signals 0
WB	No operation	All signals 0

Execution Flow

IF → ID → EX

Jump completes the early because the PC is updated in the ID/EX stage, so EX, MEM, and WB perform no useful role.

5.8 Store Word

SW stores a registers content into memory. similar to LW but performs a memory write instead.

Stage	Operation	Control Signal
IF	Fetch Instruction	PCWrite = 1, IRWrite = 1, MemRead = 1, ALUSrc = 0, PCSource = 00, MemWrite = 0, RegDst = 0, RegWrite = 0, MemtoReg = 0, ALUSrcA = 0, ALUOp = 00, Branch = 0
D	Read rs, rt, compare branch target	PCWrite = 0, PCWriteCond = 0; PCSource = 00; I or D = 0, MemRead = 0, MemWrite = 0, IRWrite = 0, RegWrite = 0, RegDst = 0, MemtoReg = 0, ALUSrcA = 0, ALUSrcB = 11, ALUOp = 00, Branch = 0
EX	Effective Address = A + sign - extended immediate	PCWrite = 0, PCWriteCond = 0 PCSource = 0, MemRead = 0, MemWrite = 0, IRWrite = 0, I or D = 0, RegWrite = 0, RegDst = 0 MemtoReg = 0, ALUSrcA = 1, ALUSrcB = 10, ALUOp = 00

MEN

WB

Exec

SW
It a
com

MEM

Mem [ALUOut] $\leftarrow$ B

PCWrite = 0, PCWriteCond = 0
PCSource = 00, MemRead = 0
MemWrite = 1,
IRWrite = 0, RegWrite = 0,
RegDst = 0, MemtoReg = 0,
ALUSrc = 0, ALUSrcB = 00,
ALUOp = 00; Branch = 0

WB

No write back

RegWrite = 0, all other 0

Execution Flow

IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ MEM

SW does not require a write back stage because
It only stores a registers value into memory after
computing the effective address